Example 4.3

Find the values of each of the following trigonometric ratios.

(i) $\sin 150^\circ$ (ii) $\cos(-210^\circ)$
(iii) $\operatorname{cosec} 390^\circ$ (iv) $\tan(-1215^\circ)$
(v) $\sec 1485^\circ$

Solution

(i) $\sin 150^\circ = \sin(1\times 90^\circ + 60^\circ)$

Since 150° lies in the second quadrant, we have

$$\sin 150^\circ = \sin(1\times 90^\circ + 60^\circ)$$
$$= \cos 60^\circ = \frac{1}{2}$$

(ii) we have $\cos(-210^o) = \cos 210^o$

Since 210° lies in the third quadrant, we have

$$\cos 210^o = \cos(180^o + 30^o)$$
$$= -\cos 30^o = -\frac{\sqrt{3}}{2}$$

(iii) $\operatorname{cosec} 390^o = \operatorname{cosec}(360^o + 30^o)$
$= \operatorname{cosec} 30^o = 2$

(iv) $\tan(-1215^o) = -\tan(1215^o)$
$= -\tan(3\times 360^o + 135^o)$
$= -\tan 135^o$
$= -\tan(90^o + 45^o)$
$= -(-\cot 45^o) = 1$

(v) $\sec 1485^o = \sec(4 \times 360^o + 45^o)$
$= \sec 45^o = \sqrt{2}$

Example 4.4

Prove that

$\sin 600^o \cos 390^o + \cos 480^o \sin 150^o = -1$

Solution

$\sin 600^o = \sin(360^o+240^o) = \sin 240^o$

$= \sin(180^o + 60^o) = -\sin 60^o = -\frac{\sqrt{3}}{2}$

$\cos 390^o = \cos(360^o + 30^o)$
$= \cos 30^o = \frac{\sqrt{3}}{2}$

$\cos 480^o = \cos(360^o + 120)$
$= \cos 120^o$
$= \cos(180^o - 60^o) = -\cos 60^o$
$= -\frac{1}{2}$

$\sin 150^o = \sin(180^o - 30^o)$
$= \sin 30^o = \frac{1}{2}$

Now, $\sin 600^o \cos 390^o + \cos 480^o \sin 150^o$

$$= \left(-\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(-\frac{1}{2}\right)\left(\frac{1}{2}\right)$$

$$= -\frac{3}{4} - \frac{1}{4} = \frac{-3-1}{4} = -1$$

Example 4.5

Prove that

$$\frac{\sin(-\theta)\tan(90^\circ-\theta)\sec(180^\circ-\theta)}{\sin(180+\theta)\cot(360-\theta)\operatorname{cosec}(90^\circ-\theta)} = 1$$

Solution

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$$= \frac{\sin(-\theta)\tan(90^\circ-\theta)\sec(180^\circ-\theta)}{\sin(180+\theta)\cot(360-\theta)\operatorname{cosec}(90^\circ-\theta)}$$

$$= \frac{(-\sin\theta)\cot\theta(-\sec\theta)}{(-\sin\theta)(-\cot\theta)\sec\theta} = 1$$

Exercise 4.1

1. Convert the following degree measure into radian measure

(i) 60° (ii) 150°
(iii) 240° (iv) -320°

2. Find the degree measure corresponding to the following radian measure.

(i) $\frac{\pi}{8}$ (ii) $\frac{9\pi}{5}$
(iii) -3 (iv) $\frac{11\pi}{18}$

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